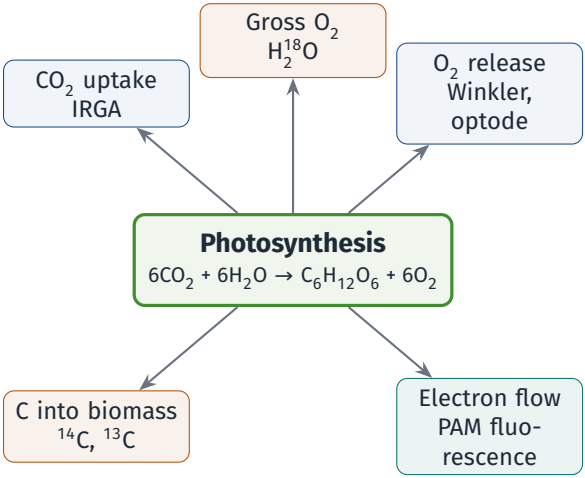
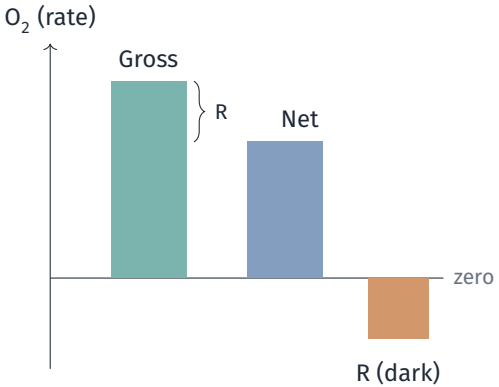
What can we measure?

- Each part of photosynthesis can be measured:
  - **CO_2 uptake**: gas exchange (IRGA)
  - **O_2 release**: Winkler titration, electrodes, optodes
  - **Carbon into biomass**: ^{14}C or ^{13}C tracers
  - **Gross O_2 production**: ^{18}O-labelled water
  - **Electron transport**: chlorophyll fluorescence (PAM)
- Methods differ in what they measure, and at what scale



Gross, net and respiration

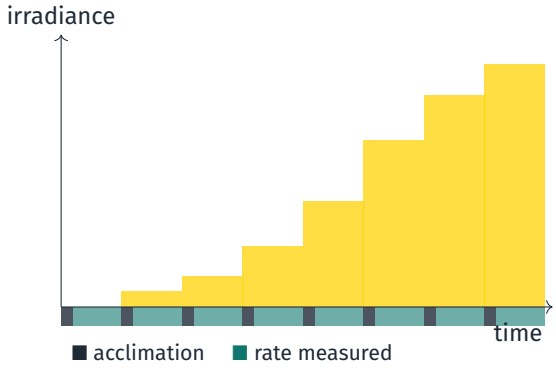
- **Net (NP)**: measured in the light; photosynthesis minus respiration
- **Respiration (R)**: measured in darkness
- **Gross**: GP = NP + R (if R is the same in light and dark)
- **PQ** = O_2 released / CO_2 fixed, about 1.0–1.4
- Normalise: per area, g mass, chlorophyll *a* or m^3 water



Schematic.

Designing a P–I experiment

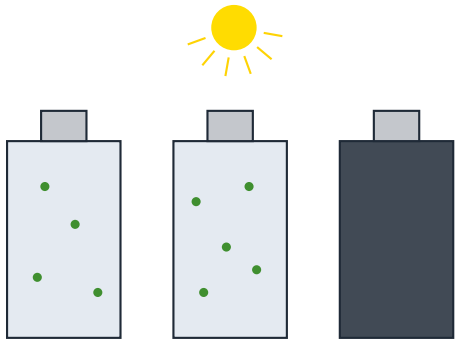
- A series of irradiances, from dark to saturating
- Each step: acclimate briefly, then measure the rate
- Hold temperature, CO_2 (DIC), O_2 and water motion constant
- Include darkness (R) and controls (no organism)
- Independent replicates; record mass, area or chlorophyll



Schematic of a stepped light series.

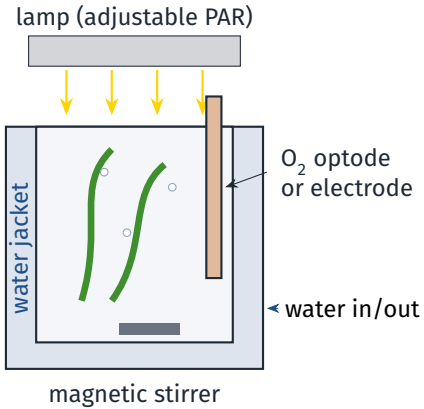
Light and dark bottles (Winkler titration)

- Fill identical bottles with the same water (or algal suspension)
- Fix one immediately (initial), incubate one clear and one dark bottle
- Measure dissolved O_2 chemically (Winkler titration) at the end
- NP = (light – initial)/*t*; R = (initial – dark)/*t*; GP = NP + R
- Example (6 h): initial 7.80, light 8.45, dark 7.55 mg O_2 L^{-1} gives GP = 0.150 mg O_2 L^{-1} h^{-1}



Oxygen sensors in a closed chamber

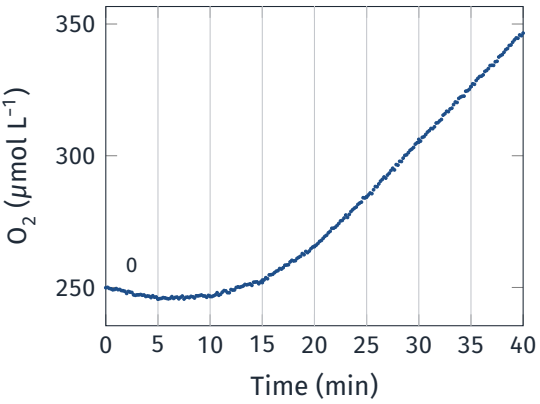
- Seaweed, seagrass, leaf discs or algae in a sealed chamber
- O_2 logged continuously: Clark electrode or optode
- Water jacket for temperature; stirrer thins the boundary layer
- Adjustable lamp; PAR measured at the sample
- Keep incubations short so O_2 and DIC change little



Schematic, not to scale.

From an oxygen trace to a rate

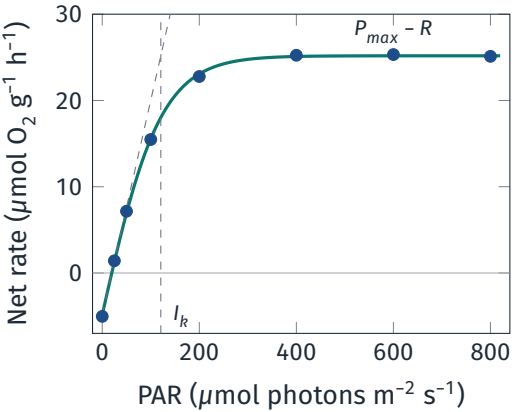
- Each light step gives a straight segment; its slope is the rate of O_2 change (μM min^{-1})
- Discard the first minute (acclimation), fit a line to the rest
- Rate = slope × 60 × volume (L) / mass (g)
- Units: μmol O_2 g^{-1} FW h^{-1}
- In darkness the slope is negative: respiration



Illustrative data: 2.0 g fresh mass of seaweed in 0.20 L, eight 5-min light steps (0–800 μmol m^{-2} s^{-1}).

The P–I curve and its parameters

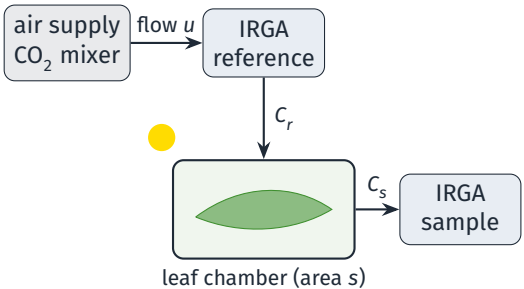
- Plot rate against irradiance; fit $P = P_{max} \tanh(\alpha I / P_{max}) - R$
- α: efficiency at low light; P_{max}: capacity
- $I_k = P_{max} / \alpha$: onset of saturation
- I_c: compensation point, where net rate = 0
- Example fit: P_{max} = 30.0, α = 0.249, R = 4.8; I_k = 121, I_c = 20



Units: P_{max} and R in μmol O_2 g^{-1} FW h^{-1}; I_k, I_c in μmol photons m^{-2} s^{-1}.

Leaf gas exchange: the open system

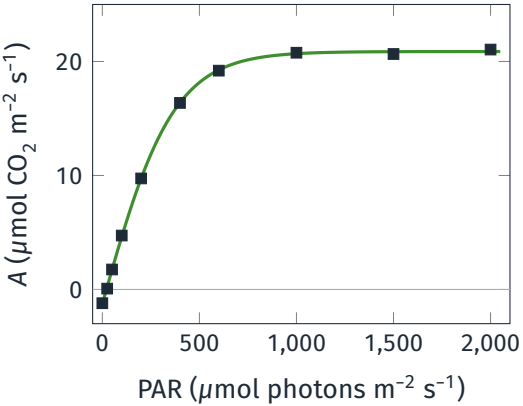
- Air of known CO_2 flows through a clamp-on leaf chamber
- Infrared gas analysers (IRGAs) measure CO_2 (and H_2O) entering (reference) and leaving (sample)
- The leaf removes CO_2, so sample CO_2 is lower
- Chamber controls light, temperature, humidity and CO_2
- Also gives transpiration and stomatal conductance



Schematic of an open (flow-through) system.

From CO_2 differences to a light response curve

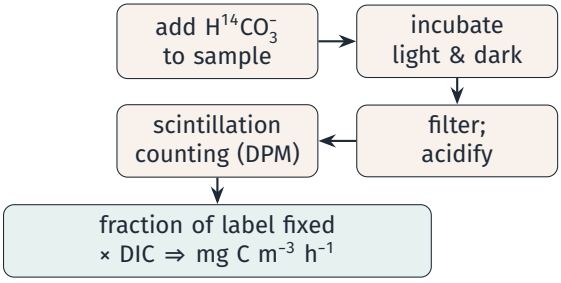
- $A = u\,(C_r - C_s)/s$ (ignoring the small water-vapour correction)
- u = 500 μmol s^{-1}, s = 6 cm^2: a difference of 25 μmol mol^{-1} means $A \approx 21$ μmol CO_2 m^{-2} s^{-1}
- Light steps usually run from high to low, with time to stabilise at each
- Same parameters as before: A_{max}, α, R_d, I_k, I_c



Illustrative leaf: fit A_{max} = 22.1, α = 0.060, R_d = 1.27 μmol CO_2 m^{-2} s^{-1}; I_k = 367, I_c = 21.

The ^{14}C method

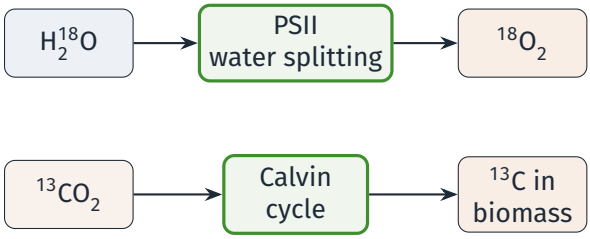
- Add a known amount of ^{14}C-bicarbonate to light and dark bottles (Steemann Nielsen 1952)
- Incubate (often 2–24 h), filter, remove unfixed ^{14}C with acid
- Count ^{14}C fixed into organic matter (scintillation counter)
- Rate: $P = \frac{DPM_{light} - DPM_{dark}}{DPM_{added}} \times \mathrm{DIC} \times 1.05/t$
- Very sensitive: the standard for phytoplankton production



Example: 5 μCi added, DIC 2.0 mmol L^{-1}, 2 h, light 3,350 and dark 150 DPM: P = 3.6 mg C m^{-3} h^{-1}.

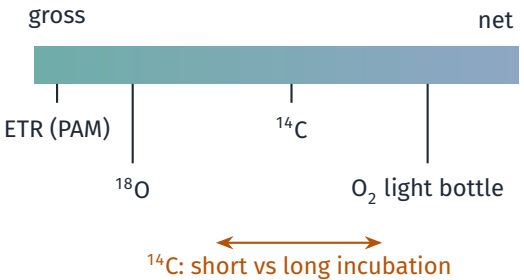
Stable isotopes: ^{18}O and ^{13}C

- **$H_2^{18}O$**: add ^{18}O-labelled water; O_2 from photosynthesis comes from water, so $^{18}O_2$ appears
- Measures **gross** O_2 production directly; respiration does not remove the label quickly
- **^{13}C**: non-radioactive alternative to ^{14}C, measured by mass spectrometry
- Need mass spectrometers; more expensive, but no radioactive waste



Which rate does each method measure?

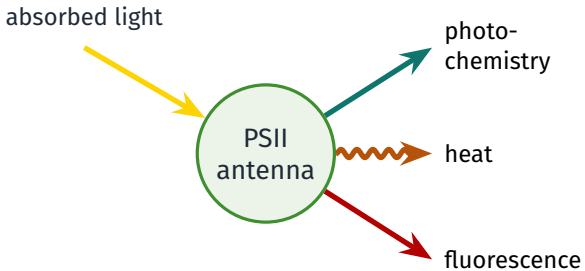
- ^{18}O: gross O_2 production
- O_2 bottles: net (gross if light R = dark R)
- ^{14}C: between gross and net, depending on incubation time (Peterson 1980)
- Fluorescence: electron transport, an upper limit to gross
- ^{14}C often 60–100% of ^{18}O gross (Grande et al. 1989)



Schematic; positions are qualitative.

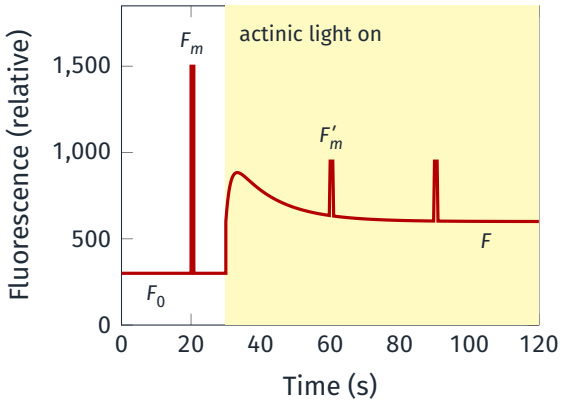
Chlorophyll fluorescence: three fates of absorbed light

- Light absorbed by PSII is used in photochemistry, lost as heat, or re-emitted as fluorescence
- The three compete: when photochemistry falls, fluorescence and heat rise
- Fluorescence can therefore report the efficiency of photochemistry
- PAM fluorometers use a modulated measuring light, so fluorescence can be measured even in sunlight



The saturating pulse

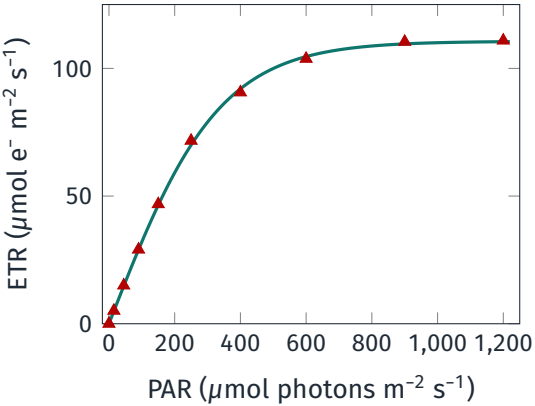
- Dark-adapted: F_0 (minimum), then a strong pulse closes all PSII centres: F_m
- $F_v/F_m = (F_m - F_0)/F_m$: maximum quantum yield (healthy leaves ≈ 0.8)
- In actinic light: steady F, and a pulse gives F'_m
- $\Phi_{PSII} = (F'_m - F)/F'_m$: effective quantum yield (Genty et al. 1989)
- Example: F = 600, F'_m = 950 gives Φ_{PSII} = 0.37



Schematic trace (values chosen for illustration).

Electron transport rate and rapid light curves

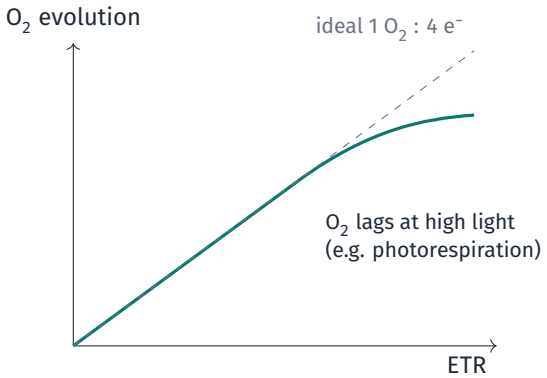
- ETR = Φ_{PSII} × PAR × 0.5 × absorptance (0.84 for many leaves)
- Rapid light curve: short (e.g. 10 s) steps of increasing light, a pulse at each
- Fit the same tanh model: ETR_{max}, α, E_k (Ralph & Gademann 2005)
- Fast and non-invasive: suits field work, seagrasses and seaweeds in situ



Illustrative rapid light curve: ETR_{max} = 111 μmol e^- m^{-2} s^{-1}, α = 0.33, E_k = 334 μmol photons m^{-2} s^{-1}.

Fluorescence: strengths and cautions

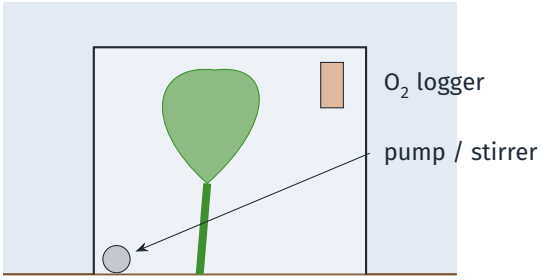
- Fast, no enclosure, works in situ (Diving-PAM)
- Counts electrons, not C or O_2: 4 e^- per O_2 in theory
- Absorptance often assumed: *relative* ETR
- Light history changes RLC parameters
- Calibrate against O_2 or ^{14}C for absolute rates (Beer et al. 1998)



Schematic, after the pattern described by Beer et al. (1998).

Seaweeds and seagrasses

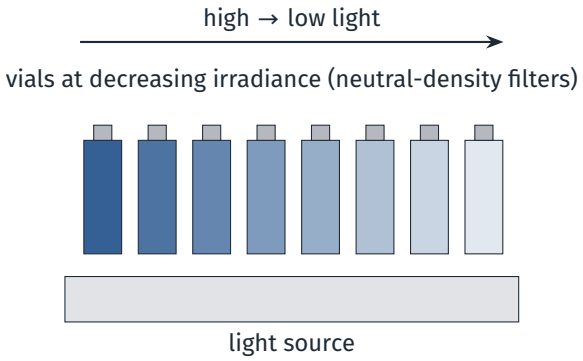
- Usually O_2 evolution in stirred, temperature-controlled chambers
- Stirring matters: a thick boundary layer limits supply of CO_2/HCO_3^-
- In situ chambers enclose whole kelp or reef communities (Rodgers et al. 2015)
- Diving-PAM for in situ ETR and rapid light curves (Ralph et al. 1998)
- Normalise to fresh or dry mass, area or chlorophyll



in situ incubation chamber (schematic)

Microalgae and phytoplankton

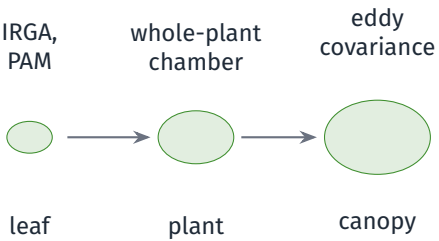
- Cells are dilute, so rates per bottle are small: sensitive methods are needed
- ^{14}C incubations in a photosynthetron (many light levels at once) give P–I curves
- O_2 light–dark bottles and ^{18}O for gross and net rates
- Normalise to chlorophyll *a*: P^B_{max}, α^B
- Active fluorescence (PAM, FRRF) for high-frequency, in situ estimates



Schematic photosynthetron.

Terrestrial plants: leaf to canopy

- Portable gas exchange: light and CO_2 responses of attached leaves (Long & Bernacchi 2003)
- Leaf-disc O_2 electrodes: capacity at high CO_2
- Fluorescence: rapid stress screening (F_v/F_m)
- Canopies: large chambers or eddy covariance towers
- A leaf rate is not a canopy rate



Choosing a method

Method	Measures	Typical use	Main limitation
Winkler bottles	net, gross (O_2)	plankton, lakes	slow; low sensitivity at low biomass
O_2 electrode/optode	net O_2 (continuous)	seaweeds, leaf discs, algae	enclosure; stirring and O_2 build-up
IRGA gas exchange	net CO_2	leaves, whole plants	air only; chamber conditions
^{14}C	C fixed (gross–net)	phytoplankton	radioactive; meaning depends on incubation time
^{18}O / ^{13}C	gross O_2 / C fixed	plankton, cultures	mass spectrometry; cost
PAM fluorescence	electron transport	everything, in situ	relative; not carbon fixation

Summary

- Photosynthesis can be followed through CO_2, O_2, carbon or electrons; each method sees a different part
- Distinguish gross, net and respiration, and state how rates are normalised
- Good design: stepped irradiance, acclimation, constant conditions, controls, replicates
- Raw data (slopes, CO_2 differences, DPM, fluorescence yields) become rates by simple equations
- Rates against irradiance give P–I curves; fitted parameters (α, P_{max}, I_k, I_c, R) let us compare species, sites and treatments
- Choose the method that matches the organism, the scale and the question